

HAYS, KS, USA

WMO: 724518

Lat: 38.850N

Lon: 99.267W

Elev: 609

StdP: 94.22

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03968

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.2	-13.4	-21.4	0.6	-13.5	-18.7	0.8	-11.5	14.5	5.4	12.7	2.2	4.0	340	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	38.3	21.9	37.1	21.9	34.9	21.7	24.2	32.7	23.5	32.3	22.8	31.8	7.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.1	18.1	28.1	21.2	17.1	27.2	20.2	16.0	26.5	76.2	32.9	73.0	32.2	70.4	31.9	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.8	10.7	-20.0	41.3	2.3	1.6	-21.6	42.5	-23.0	43.4	-24.3	44.3	-25.9	45.5		
(5)			-20.4	25.5	2.1	0.9	-22.0	26.1	-23.2	26.6	-24.4	27.1	-25.9	27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	-0.9	0.8	6.4	11.4	17.3	23.6	26.4	25.2	20.7	13.0	5.8	-0.3		
	DBStd	10.99	5.97	6.57	6.17	5.38	4.84	4.10	3.50	3.37	4.88	5.32	5.26	5.58		
	HDD10.0	1286	338	262	144	45	3	0	0	0	1	29	144	319		
	HDD18.3	2920	596	491	372	214	78	6	1	1	31	178	376	577		
	CDD10.0	2207	1	5	32	88	230	409	508	471	322	123	18	1		
	CDD18.3	797	0	0	1	7	47	166	250	213	101	13	0	0		
	CDH23.3	9681	1	3	33	154	608	2013	3106	2401	1158	196	9	0		
	CDH26.7	4998	0	1	4	47	244	1043	1776	1276	544	63	1	0		
(14) Wind	WSAvg	4.9	4.4	4.9	5.4	5.9	5.2	5.3	4.7	4.4	4.9	5.0	4.7	4.4		
(15) Precipitation	PrecAvg	577	11	16	43	50	75	84	94	72	55	36	25	14		
	PrecMax	965	42	69	201	134	228	196	294	180	217	96	94	54		
	PrecMin	363	0	0	0	1	4	18	0	6	6	0	0	0		
	PrecStd	129	11	16	43	29	43	50	59	46	43	27	26	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	22.5	27.0	31.2	36.0	40.2	41.2	39.1	37.6	32.3	24.4	17.9		
		MCWB	8.3	10.7	14.2	15.8	19.1	21.3	21.8	21.6	20.7	18.4	13.1	9.4		
	2%	DB	14.8	18.0	23.0	27.6	32.2	37.2	38.9	37.5	34.6	27.9	21.2	14.1		
		MCWB	6.5	9.0	12.7	15.5	19.4	21.6	22.2	22.1	20.8	16.5	11.8	7.2		
	5%	DB	11.8	14.1	20.5	24.1	29.0	34.7	37.4	35.9	32.4	25.2	17.8	12.0		
		MCWB	5.2	7.2	11.6	14.5	18.6	21.6	22.3	22.1	20.5	15.6	10.2	6.1		
	10%	DB	8.0	11.3	17.3	21.4	27.1	32.5	35.1	33.0	29.9	22.4	15.1	8.5		
		MCWB	3.3	5.7	10.4	13.2	18.0	21.1	22.2	21.7	19.4	14.5	8.9	4.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	11.9	17.0	18.5	22.4	24.7	25.4	25.1	23.4	20.8	15.0	10.9		
		MCDB	17.1	18.8	23.1	26.0	29.6	33.6	34.3	32.8	31.5	27.9	21.1	14.8		
	2%	WB	7.1	9.9	14.9	16.9	21.2	23.6	24.2	24.0	22.2	18.6	12.8	7.9		
		MCDB	14.0	17.1	19.4	24.2	28.6	32.5	33.0	32.2	31.1	24.7	18.5	13.4		
	5%	WB	5.4	7.8	12.7	15.4	20.0	22.5	23.5	23.2	21.3	17.0	11.0	6.0		
		MCDB	11.1	13.9	18.8	22.6	26.7	31.8	32.7	31.9	29.8	23.2	17.0	11.5		
	10%	WB	3.7	5.8	10.5	13.9	18.8	21.7	22.9	22.4	20.3	15.2	9.3	4.1		
		MCDB	8.0	11.1	17.3	20.9	25.2	30.8	32.3	31.0	27.8	21.2	15.0	8.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	13.4	14.4	14.3	13.7	14.2	14.2	13.7	14.1	14.3	14.1	12.4		
		MCDBR	18.9	19.4	19.8	19.4	17.6	17.1	17.3	17.0	16.7	18.6	18.9	17.5		
		MCWBR	12.2	12.0	11.2	10.0	8.0	6.2	5.6	5.9	6.6	9.2	11.2	11.5		
		MCDBR	17.9	18.4	16.9	16.6	14.9	15.2	14.5	14.3	14.5	15.0	17.5	16.6		
	5% WB	MCWBR	11.8	11.7	10.3	9.4	8.0	6.6	5.7	6.0	6.5	8.4	10.7	11.3		
(40) Clear-Sky Solar Irradiance	taub		0.271	0.279	0.312	0.339	0.364	0.385	0.407	0.392	0.356	0.312	0.278	0.266		
	taud		2.532	2.514	2.439	2.392	2.404	2.377	2.351	2.386	2.422	2.532	2.550	2.538		
	Ebn at Noon		923	958	950	937	914	892	869	875	889	906	905	902		
	Edh at Noon		74	86	104	116	117	121	123	116	105	84	72	68		
(44) All-Sky Solar Radiation	RadAvg		2.49	3.30	4.46	5.52	6.36	7.11	6.95	6.13	5.08	3.70	2.76	2.18		
	RadStd		0.20	0.33	0.36	0.30	0.41	0.39	0.31	0.37	0.30	0.39	0.23	0.18		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page